

Ask VJ

Data Warehouse Architecture Technical Reference Manual

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Chapter 1: Architecture Overview

1.1 Medallion Architecture

Ask VJ uses a three-layer medallion architecture (Bronze > Silver > Gold) to unify customer, booking, and sales data from three independent source systems into a single business-ready data warehouse. An AI layer sits on top of Gold, generating SQL from natural language questions.

1.2 Source Systems

System	Database	Type	Key Identifier
Farvision ERP	VJDLIVE45	SQL Server	TenantId = 75
VJ Sales App	PostgreSQL	Supabase	UUID IDs
VJOP Portal	RefferalAndLoyalty	SQL Server	INT IDs

1.3 Universal Project Key: BUIId

BUIId (BusinessUnitId) is the universal project identifier across all three systems:

- Farvision: ENGG.DimBusinessUnit.BusinessUnitId
- VJ Sales: Projects.buld
- VJOP: customer_bookings_units.BUIId

All cross-system project joins resolve through BUIId via silver.project_crosswalk.

1.4 Cross-System Join Keys

Join Key	Farvision Column	Other System	Confidence
BookingId	DimBookingMaster.BookingId	VJ Sales: bookingId	0.99
		VJOP: fv_booking_id	0.99
UnitId	DimUnitMaster.UnitId	VJ Sales: farvisionUnitId	0.95
		VJOP: farvisionUnitId	0.95
BUIId	DimBusinessUnit.BusinessUnitId	VJ Sales: buld	1.00
		VJOP: BUIId	1.00
LedgerId	DimCustomerDetail.LedgerCustId	VJOP: FV_LedgerID	0.95
LeadId	(none)	VJ Sales: leadId <> VJOP: sales_app_leadId	0.95

1.5 Architecture Diagram

SOURCE SYSTEMS	BRONZE	SILVER	GOLD	AI LAYER
+-----+	+-----+	+-----+	+-----+	+-----+
Farvision ERP	Raw	Entity	Star	LLM SQL
(SQL Server)	Staging	Resolution	Schema	Generation
+-----+	16 tables		7 dims	
VJ Sales App		ID-based	4 facts	Natural
(PostgreSQL)	13 tables	Matching	3 snaps	Language
+-----+			1 view	Answers
VJOP Portal	7 tables	Fuzzy		
(SQL Server)		Fallback		
+-----+	+-----+	+-----+	+-----+	+-----+

Chapter 2: Bronze Layer -- Raw Staging

The Bronze layer contains 36 staging tables: 16 from Farvision ERP, 13 from VJ Sales App, and 7 from VJOP. All tables are append-only with audit metadata: `_sync_id` (BIGSERIAL), `_synced_at` (TIMESTAMPTZ), `_source_system` (VARCHAR), `_batch_id` (UUID). No transformations are applied -- data is preserved exactly as received from source systems.

2.1 Farvision ERP (16 Tables)

All Farvision queries include WHERE TenantId = 75. Column names are PascalCase (quoted).

Table: bronze.stg_fv_dim_booking_master

Source: CRMG.DimBookingMaster

Column	Type	Description
BookingId	INT PK	Booking identifier
LedgerId	INT	Customer ledger ID (financial)
FiscalYearId	INT	56=FY2025-26, 52=FY2024-25
BUID	INT	Universal project key
BookingDate	DATE	When booking was made
BookingNo	VARCHAR	Booking reference number
PrimaryUnitId	INT	Main unit in booking
CustomerName	VARCHAR	Customer display name
NetBasicPrice	DECIMAL	Base price before taxes
SalesPersonId	INT	Sales person reference
SalesPersonName	VARCHAR	Sales person name
IsCancelled	BOOLEAN	0=active, 1=cancelled
DiscountPercentage	DECIMAL	Discount applied
AllotmentDate	DATE	Unit allotment date
AgreementDate	DATE	Agreement signing date
AgreementNo	VARCHAR	Agreement reference
RegistrationDate	DATE	Legal registration date
RegistrationNo	VARCHAR	Registration reference

Table: bronze.stg_fv_fact_unit_movement

Source: CRMG.FactUnitMovement

Column	Type	Description
UnitId	INT	Unit identifier
UnitStatus	INT	1=Sold, 2=Available, 3=Blocked
BUID	INT	Project key
TypologyId	INT	Unit type (1BHK, 2BHK, etc)
Area	DECIMAL	Saleable area
CarpetArea	DECIMAL	Net usable area
BookingId	INT	Linked booking
Value	DECIMAL	Total unit value

Table: bronze.stg_fv_dim_unit_master

Source: CRMG.DimUnitMaster

Column	Type	Description
UnitId	INT PK	Unit identifier
UnitCode	VARCHAR	Unit code/number
BUIId	INT	Project key
TypologyId	INT	Unit type reference
FloorId	INT	Floor reference

Table: bronze.stg_fv_dim_typology_master

Source: CRMG.DimTypologyMaster

Column	Type	Description
TypologyId	INT PK	Typology identifier
TypologyCode	VARCHAR	Code e.g. 3BHK-XL
Typology	VARCHAR	Name e.g. 3.00BHK

Table: bronze.stg_fv_dim_receipt

Source: CRMG.DimReceipt

Column	Type	Description
ReceiptId	INT PK	Receipt ID (sic spelling)
BookingId	INT	Linked booking
AmountLCY	DECIMAL	Amount in local currency
PaymentMode	VARCHAR	Cheque/NEFT/RTGS/Cash
InstrumentNo	VARCHAR	Cheque/NEFT reference
DocumentNo	VARCHAR	Document reference
DocumentDate	DATE	Payment date
LedgerId	INT	Customer ledger

Table: bronze.stg_fv_dim_invoice

Source: CRMG.DimInvoice

Column	Type	Description
InvoiceId	INT PK	Invoice identifier
BookingId	INT	Linked booking
BasicAmount	DECIMAL	Base invoice amount
AmountLCY	DECIMAL	Total invoice amount
DocumentDate	DATE	Invoice date

Table: bronze.stg_fv_fact_duedate_outstanding

Source: CRMG.FactDueDatewiseOutstanding

Column	Type	Description
LedgerId	INT	Customer ledger
CustomerName	VARCHAR	Customer name
UnitNo	VARCHAR	Unit number
Bill_Amount	DECIMAL	Total billed
Paid_Amount	DECIMAL	Total paid
Due_Amount	DECIMAL	Outstanding due
OverdueDays	INT	Days overdue
DayAmt_15 to DayAmt_More180	DECIMAL	7 aging buckets
Buld	INT	Project key

Table: bronze.stg_fv_dim_customer_detail

Source: CRMG.DimCustomerDetail

Column	Type	Description
LedgerCustId	INT PK	Customer/ledger ID
Customer	VARCHAR	Display name
FullName	VARCHAR	Full legal name
MobileNo	VARCHAR	Mobile number
EmailId	VARCHAR	Email address
PanNo	VARCHAR	PAN number

Table: bronze.stg_fv_dim_business_unit

Source: ENGG.DimBusinessUnit

Column	Type	Description
BusinessUnitId	INT PK	Universal project key (BUId)
BusinessUnit	VARCHAR	Project name
BusinessUnitType	VARCHAR	Type classification
SegmentId	INT	Segment reference

2.2 VJ Sales App (13 Tables)

Timestamps are EPOCH MILLISECONDS (BIGINT). IDs are UUIDs. JSONB preserved as-is.

Table: bronze.stg_vj_projects

Source: "Projects"

Column	Type	Description
projectId	UUID PK	Project identifier
projectName	VARCHAR	Project name
buld	INT	Maps to Farvision BUId
reraNumber	VARCHAR	RERA registration
isCompleted	BOOLEAN	Project completed flag

Table: bronze.stg_vj_inventory

Source: "Inventory"

Column	Type	Description
unitId	UUID PK	Unit identifier
projectId	UUID	Project reference
farvisionUnitId	INT	Cross-system key to Farvision
totalCost	DECIMAL	Total flat price
BSP	DECIMAL	Base selling price/sqft
saleableArea	DECIMAL	Saleable area
inventoryStatusId	INT	Status lookup

Table: bronze.stg_vj_leads

Source: "Leads"

Column	Type	Description
leadId	UUID PK	Lead identifier
personId	UUID	Person reference
cpId	UUID	Channel partner ref
leadType	VARCHAR	walk-in/referral/digital
created_at	BIGINT	Epoch milliseconds

Table: bronze.stg_vj_allotment_payment

Source: "AllotmentPayment"

Column	Type	Description
allotmentPaymentId	UUID PK	Allotment identifier
leadId	UUID	Lead reference
unitId	UUID	Unit reference
bookingId	INT	Cross-system key to Farvision
status	VARCHAR	Payment Complete/Agreement Done
paidAmount	DECIMAL	Amount paid
bookingAmt	JSONB	Booking amount details
unitCost	JSONB	Unit cost details

Table: bronze.stg_vj_person

Source: "Person"

Column	Type	Description
personId	UUID PK	Person identifier
name	VARCHAR	Full name
contactNumber	VARCHAR	Phone number
email	VARCHAR	Email address

2.3 VJOP Referral & Loyalty (7 Tables)**Table: bronze.stg_rnl_customers**Source: *dbo.customers*

Column	Type	Description
id	INT PK	VJOP customer ID
name	VARCHAR	Customer name
mobile	VARCHAR	Mobile number
FV_LedgerID	INT	Cross-system key to Farvision LedgerId
PAN	VARCHAR	PAN number

Table: bronze.stg_rnl_customer_bookings_unitsSource: *dbo.customer_bookings_units*

Column	Type	Description
id	INT PK	Record ID
customer_id	INT	VJOP customer ref
fv_booking_id	INT	Cross-system key to Farvision BookingId
BUIId	INT	Cross-system key to Farvision BUIId
project	VARCHAR	Project name

Table: bronze.stg_rnl_leadsSource: *dbo.leads*

Column	Type	Description
id	INT PK	VJOP lead ID
name	VARCHAR	Lead name
sales_app_lead_id	UUID	Cross-system key to VJ Sales leadId
referred_by	INT	Referrer customer ID
status	VARCHAR	Referral status
booking_id	INT	Booking reference

Chapter 3: Silver Layer -- Entity Resolution

The Silver layer resolves cross-system identities using shared IDs. ID-based matching is the PRIMARY strategy. Fuzzy matching (pg_trgm) is only used as a fallback for unconverted leads with no BookingId/UnitId.

3.1 silver.project_crosswalk

Maps BUID (universal project key) to project names across all 3 systems.

Column	Type	Description
crosswalk_id	SERIAL PK	Auto-increment ID
canonical_name	VARCHAR	Unified project name for Gold
farvision_bu_id	INT UNIQUE	Farvision BusinessUnitId (universal key)
vjsales_project_name	VARCHAR	Name from VJ Sales Projects table
vjop_project_name	VARCHAR	Name from VJOP customer_bookings_units
phase_name	VARCHAR	Phase within project

3.2 silver.entity_map

Links customer/booking identities across systems. Each row = one resolved identity.

Column	Type	Description
entity_map_id	SERIAL PK	Auto-increment ID
unified_customer_id	UUID	System-generated unified ID
farvision_booking_id	INT UNIQUE	DimBookingMaster.BookingId
farvision_unit_id	INT	DimUnitMaster.UnitId
farvision_bu_id	INT	DimBusinessUnit.BusinessUnitId
farvision_ledger_id	INT	DimCustomerDetail.LedgerCustId
vjsales_lead_id	UUID UNIQUE	Leads.leadId
vjsales_allotment_id	UUID UNIQUE	AllotmentPayment.allotmentPaymentId
vjsales_unit_id	UUID	Inventory.unitId
vjop_customer_id	INT	customers.id
vjop_lead_id	INT	leads.id
match_method	VARCHAR	booking_id_exact/unit_id_exact/lead_id_exact/fuzzy
match_confidence	DECIMAL	0.00-1.00 (0.99 for booking match)

3.3 Resolution Strategy (Priority Order)

```

Strategy 1: BookingId Exact Match (confidence: 0.99)
  Farvision BookingId <-> VJ Sales bookingId <-> VJOP fv_booking_id
  JOIN: stg_fv_dim_booking_master.BookingId = stg_vj_allotment_payment.bookingId
        stg_fv_dim_booking_master.BookingId = stg_rnl_customer_bookings_units.fv_booking_id

Strategy 2: UnitId Exact Match (confidence: 0.95)
  Farvision UnitId <-> VJ Sales farvisionUnitId
  JOIN: stg_vj_inventory.farvisionUnitId = (Farvision UnitId)

Strategy 3: LeadId Exact Match (confidence: 0.95)
  VJ Sales leadId <-> VJOP sales_app_lead_id
  JOIN: stg_rnl_leads.sales_app_lead_id = stg_vj_leads.leadId

Strategy 4: Fuzzy Fallback (queue only)
  For unconverted leads with no BookingId/UnitId
  Queued in silver.entity_resolution_queue for manual review

```


Chapter 4: Gold Layer -- Star Schema

The Gold layer provides business-ready tables optimized for LLM SQL generation. 7 dimension tables, 4 append-only fact tables, 3 daily snapshot tables, and 1 computed view. All use surrogate keys for simple JOINS.

4.1 Dimension Tables

Table: gold.dim_date

Grain: One row per date | Source: Generated (2015-2035) | Refresh: One-time

Column	Type	Description
date_key	INT PK	YYYYMMDD format
full_date	DATE	Actual date value
fiscal_quarter	VARCHAR	Q1=Apr-Jun, Q2=Jul-Sep, Q3=Oct-Dec, Q4=Jan-Mar
fiscal_year	INT	Indian FY (2026 = Apr 2025 - Mar 2026)
fiscal_year_id	INT	Farvision ID: 56=FY2025-26, 52=FY2024-25
is_weekend	BOOLEAN	Saturday or Sunday

Table: gold.dim_projects

Grain: One row per project | Source: silver.project_crosswalk | Refresh: Upsert on bu_id

Column	Type	Description
project_key	SERIAL PK	Surrogate key
bu_id	INT UNIQUE	Farvision BusinessUnitId (universal)
project_name	VARCHAR	Canonical project name
phase_name	VARCHAR	Phase within project
segment	VARCHAR	Residential/Commercial
status	VARCHAR	active/completed/upcoming

Table: gold.dim_typologies

Grain: One row per typology variant | Source: stg_fv_dim_typology_master | Refresh: Upsert

Column	Type	Description
typology_key	SERIAL PK	Surrogate key
typology_id	INT	Farvision TypologyId
display_name	VARCHAR	Human-friendly: 3 BHK
is_base_variant	BOOLEAN	true=base, false=XL/XR

Table: gold.dim_units

Grain: One row per unit | Source: stg_fv_dim_unit_master + FactUnitMovement | Refresh: Upsert

Column	Type	Description
unit_key	SERIAL PK	Surrogate key
farvision_unit_id	INT UNIQUE	Farvision UnitId
project_key	INT FK	-> dim_projects
typology_key	INT FK	-> dim_typologies
unit_no	VARCHAR	Unit number e.g. A-501
unit_status	INT	1=sold, 2=available, 3=blocked
carpet_area	DECIMAL	Net usable area sqft
saleable_area	DECIMAL	Total saleable area

Table: gold.dim_customers

Grain: One row per customer | Source: stg_fv_dim_customer_detail + entity_map | Refresh: Insert

Column	Type	Description
customer_key	SERIAL PK	Surrogate key
unified_customer_id	UUID UNIQUE	From silver.entity_map
farvision_ledger_id	INT	Farvision LedgerCustId
full_name	VARCHAR	Legal name from Farvision
mobile	VARCHAR	Mobile number
email	VARCHAR	Email address
pan_number	VARCHAR	PAN number

Table: gold.dim_sales_persons

Grain: One row per sales person | Source: stg_fv_dim_booking_master | Refresh: Upsert

Column	Type	Description
sales_person_key	SERIAL PK	Surrogate key
farvision_sales_person_id	INT UNIQUE	Farvision SalesPersonId
name	VARCHAR	Sales person name

Table: gold.dim_lead_sources

Grain: One row per source | Source: stg_vj_leads + stg_vj_cp | Refresh: Upsert

Column	Type	Description
source_key	SERIAL PK	Surrogate key
source_name	VARCHAR UNIQUE	walk-in/referral/digital-fb/CP-Name
source_category	VARCHAR	organic/paid/referral/channel_partner
cp_id	UUID	Channel partner UUID (if applicable)

4.2 Fact Tables (Append-Only / Upsert)**Table: gold.fact_bookings**

Grain: One row per booking | Source: stg_fv_dim_booking_master | Refresh: Upsert on farvision_booking_id

Column	Type	Description
booking_key	SERIAL PK	Surrogate key
farvision_booking_id	INT UNIQUE	Farvision BookingId
booking_date	DATE	Booking date
customer_key	INT FK	-> dim_customers (via entity_map)
project_key	INT FK	-> dim_projects (via BUIId)
unit_key	INT FK	-> dim_units (via PrimaryUnitId)
typology_key	INT FK	-> dim_typologies (via FactUnitMovement)
net_basic_price	DECIMAL	Base price before taxes
agreement_value	DECIMAL	Total agreement value
is_cancelled	BOOLEAN	false=active, true=cancelled
cancellation_date	DATE	From DimBookingCancellation
agreement_date	DATE	From DimUnitAgreement
registration_date	DATE	From DimUnitAgreement
broker_id	INT	From FactSalesDetailWise
fiscal_year_id	INT	56=FY2025-26

Table: gold.fact_receipts

Grain: One row per payment | Source: stg_fv_dim_receipt | Refresh: Upsert on farvision_receipt_id

Column	Type	Description
receipt_key	SERIAL PK	Surrogate key
farvision_receipt_id	INT UNIQUE	Farvision ReceiptId (sic)
booking_key	INT FK	-> fact_bookings (via BookingId)
amount	DECIMAL	Payment amount
payment_mode	VARCHAR	Cheque/NEFT/RTGS/Cash
date_key	INT FK	-> dim_date (from DocumentDate)

Table: gold.fact_invoices

Grain: One row per invoice | Source: stg_fv_dim_invoice | Refresh: Upsert on farvision_invoice_id

Column	Type	Description
invoice_key	SERIAL PK	Surrogate key
farvision_invoice_id	INT UNIQUE	Farvision InvoiceId
booking_key	INT FK	-> fact_bookings (via BookingId)
basic_amount	DECIMAL	Base invoice amount
total_amount	DECIMAL	Total with taxes
due_date	DATE	Payment due date

Table: gold.fact_lead_pipeline

Grain: One row per stage change | Source: stg_vj_leads/visits/allotments | Refresh: Append-only

Column	Type	Description
pipeline_event_id	SERIAL PK	Surrogate key
vjsales_lead_id	UUID	VJ Sales lead UUID
pipeline_stage	VARCHAR	inquiry/site_visit/booking/agreement/cancelled
allotment_status	VARCHAR	Payment Complete/Agreement Done/etc
event_date_key	INT FK	-> dim_date
project_key	INT FK	-> dim_projects (via buld)

4.3 Snapshot Tables (Daily Refresh)**Table: gold.snapshot_outstanding**

Grain: One row per (customer, unit, due_date) | Source: stg_fv_fact_duedate_outstanding | Refresh: DELETE+INSERT daily

Column	Type	Description
snapshot_date	DATE	Snapshot date (CURRENT_DATE)
customer_name	VARCHAR	Denormalized for fast queries
bill_amount	DECIMAL	Total billed
paid_amount	DECIMAL	Total paid
due_amount	DECIMAL	Outstanding balance
overdue_days	INT	Days past due
day_amt_15..more_180	DECIMAL	7 aging buckets

Table: gold.snapshot_inventory

Grain: One row per unit | Source: stg_vj_inventory | Refresh: DELETE+INSERT daily

Column	Type	Description
snapshot_date	DATE	Snapshot date
inventory_status	VARCHAR	Available/On Hold/Sold
total_cost	DECIMAL	Total flat price
bsp	DECIMAL	Base selling price per sqft
saleable_area	DECIMAL	Saleable area

Table: gold.snapshot_referrals

Grain: One row per referral lead | Source: stg_rnl_leads + allotments + points | Refresh: DELETE+INSERT daily

Column	Type	Description
snapshot_date	DATE	Snapshot date
referrer_customer_key	INT FK	-> dim_customers (who referred)
referred_lead_name	VARCHAR	Name of referred person
referral_status	VARCHAR	Unclaimed/Claimed/Site Visit Done/Agreement Done
booking_id	INT	From lead_allotments (NOT leads)
points_earned	DECIMAL	Loyalty points earned
points_redeemed	DECIMAL	Loyalty points redeemed

4.4 View: gold.v_conversion_rates

Derived conversion metrics computed on-the-fly from fact_daily_funnel_snapshot. Not stored (Medallion principle: no derived metrics in facts).

```

inquiry_to_visit_rate = total_site_visits / total_inquiries * 100
visit_to_booking_rate = total_bookings / total_site_visits * 100
booking_to_agreement_rate = total_agreements / total_bookings * 100

```

Chapter 5: ETL Pipeline Orchestration

Pipeline runs every 2 hours via cron. Execution: `python -m backend.etl.run_pipeline`

5.1 Pipeline Stages

Stage 1: EXTRACT (Bronze)

- Farvision: 16 full-load queries (TenantId=75)
- VJ Sales: 12 incremental queries (watermark on timestamp)
- VJOP: 6 incremental queries (watermark on created_at)

Stage 2: RESOLVE (Silver)

- Auto-discover projects from `stg_fv_dim_business_unit`
- Match VJ Sales projects via `buId`
- Run entity resolution: `BookingId -> UnitId -> LeadId -> Fuzzy queue`

Stage 2.5: PRE-GOLD QUALITY GATE

- Check `silver.project_crosswalk` is populated
- If errors: BLOCK Gold transforms, return early

Stage 3: TRANSFORM (Gold)

- Build 7 dimensions (in dependency order)
- Build 4 facts + 3 snapshots
- Build daily funnel snapshot

Stage 4: POST-GOLD VALIDATE

- Orphan record detection
- Null/negative checks
- Entity resolution coverage
- Booking completeness

5.2 Dimension Build Order (Dependencies)

1. `dim_date` (independent - generated)
2. `dim_projects` (reads: `silver.project_crosswalk`)
3. `dim_typologies` (reads: `stg_fv_dim_typology_master`)
4. `dim_units` (REQUIRES: `dim_projects`, `dim_typologies`)
5. `dim_customers` (reads: `stg_fv_dim_customer_detail`)
6. `dim_sales_persons` (reads: `stg_fv_dim_booking_master`)
7. `dim_lead_sources` (reads: `stg_vj_leads` + `stg_vj_cp`)

Chapter 6: Data Lineage Maps

6.1 Booking Lineage

```
Farvision CRMG.DimBookingMaster
-> bronze.stg_fv_dim_booking_master (raw copy)
-> silver.entity_map (matched on BookingId, confidence 0.99)
-> gold.fact_bookings (via farvision_booking_id)
-> gold.fact_receipts (via booking_key FK)
-> gold.fact_invoices (via booking_key FK)
```

Agreement/Registration dates come from:
stg_fv_dim_unit_agreement (LATERAL join on BookingId)

Cancellation date comes from:
stg_fv_dim_booking_cancellation (LATERAL join on BookingId)

Broker ID comes from:
stg_fv_fact_sales_detail_wise (LATERAL join on BookingId)

6.2 Customer Lineage

```
Farvision CRMG.DimCustomerDetail
-> bronze.stg_fv_dim_customer_detail
-> gold.dim_customers (LedgerCustId -> farvision_ledger_id)
```

VJ Sales 'Person' + 'Leads'

```
-> bronze.stg_vj_person + stg_vj_leads
-> silver.entity_map (vjsales_lead_id)
-> gold.dim_customers (unified_customer_id)
```

VJOP dbo.customers

```
-> bronze.stg_rnl_customers
-> silver.entity_map (FV_LedgerID -> farvision_ledger_id)
```

6.3 Inventory Lineage

VJ Sales 'Inventory'

```
-> bronze.stg_vj_inventory
-> gold.snapshot_inventory (farvisionUnitId -> dim_units)
-> gold.dim_units (farvisionUnitId -> farvision_unit_id)
```

Farvision CRMG.DimUnitMaster + FactUnitMovement

```
-> bronze.stg_fv_dim_unit_master + stg_fv_fact_unit_movement
-> gold.dim_units (UnitId -> farvision_unit_id)
Unit status (1=sold/2=available/3=blocked) from FactUnitMovement
```

6.4 Outstanding Lineage

```
Farvision CRMG.FactDueDatewiseOutstanding
-> bronze.stg_fv_fact_duedate_outstanding
-> gold.snapshot_outstanding (BuId -> dim_projects.bu_id)
7 aging buckets: DayAmt_15 through DayAmt_More180
```

6.5 Referral Lineage

VJOP dbo.leads

```
-> bronze.stg_rnl_leads
-> gold.snapshot_referrals
Referrer: referred_by -> stg_rnl_customers -> FV_LedgerID -> dim_customers
```

Booking: LATERAL stg_rnl_lead_allotments.booking_id

Points: LATERAL SUM(stg_rnl_points_history.points) credit/debit

Chapter 7: Business Rules Reference

7.1 Indian Fiscal Year

April-March cycle. FY2026 = April 2025 to March 2026.

Fiscal quarters: Q1=Apr-Jun, Q2=Jul-Sep, Q3=Oct-Dec, Q4=Jan-Mar.

FiscalYearId mapping: 56 = FY2025-26, 52 = FY2024-25.

7.2 Unit Status Codes

Code	Status	Description
1	Sold	Unit has been booked/sold
2	Available	Unit is available for sale
3	Blocked	Unit is reserved/blocked

7.3 Typology Classification

is_base_variant = true for base types (1BHK, 2BHK, 3BHK).

is_base_variant = false for variants (3BHK XL, 3BHK XR, DUPLEX, PENTHOUSE).

For '3 BHK' queries: use WHERE is_base_variant = true AND display_name LIKE '3 BHK%'.

7.4 Pipeline Stages (VJ Sales)

inquiry -> site_visit -> negotiation -> booking -> agreement -> registered | cancelled

7.5 Allotment Status Values

Payment Pending, Partial Payment Done, Payment Complete, Booked, Agreement Done, Cancelled, Refund Initiated, Refund Processed.

7.6 Aging Buckets (Outstanding)

Column	Range	Description
day_amt_15	0-15 days	Recently overdue
day_amt_30	16-30 days	One month overdue
day_amt_60	31-60 days	Two months overdue
day_amt_90	61-90 days	Three months overdue
day_amt_120	91-120 days	Four months overdue
day_amt_180	121-180 days	Six months overdue
day_amt_more_180	>180 days	Severely overdue

7.7 Timestamp Handling

VJ Sales App stores created_at and last_updated_at as EPOCH MILLISECONDS (not seconds). Conversion: to_timestamp(created_at / 1000). In the Gold layer, all timestamps are proper TIMESTAMP columns.

7.8 TenantId Filter

ALL Farvision ERP queries MUST include WHERE TenantId = 75. This is hardcoded as FARVISION_TENANT_ID = 75 in the transformer module.

Chapter 8: Cross-System Join Key Reference

Key	Farvision	Other System	Confidence
BookingId	DimBookingMaster.BookingId	VJ Sales: AllotmentPayment.bookingId	0.99
BookingId	DimBookingMaster.BookingId	VJOP: customer_bookings_units.fv_booking_	0.99
UnitId	DimUnitMaster.UnitId	VJ Sales: Inventory.farvisionUnitId	0.95
UnitId	DimUnitMaster.UnitId	VJOP: lead_allotments.farvisionUnitId	0.95
BUIId	DimBusinessUnit.BusinessUnitId	VJ Sales: Projects.buld	1.00
BUIId	DimBusinessUnit.BusinessUnitId	VJOP: customer_bookings_units.BUIId	1.00
LedgerId	DimCustomerDetail.LedgerCustId	VJOP: customers.FV_LedgerID	0.95
LeadId	(none)	VJ Sales leadId <-> VJOP sales_app_lead_id	0.95